

ZAFEER RANGOONWALA

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EDUCATION

The University of Texas at Dallas

Master of Science, Computer and Information Science (GPA: 3.5)

- **Coursework:** Big Data, Machine Learning, Database Management System, AI

Aug 2024 - May 2026

Richardson, TX

Indus University

Bachelor of Science, Computer and Information Science (GPA: 3.5)

- **Coursework:** Data Structures and Algorithm, , Web Development, Cryptography and Network Security, NLP

Oct 2020 - May 2024

Ahmedabad, India

SKILLS

- **Languages:** Python, SQL, Java, Go, PySpark, C/C++, JavaScript, TypeScript
- **Frameworks:** Django, Flask, PyTorch, React, Next.js, Node.js, Spring Boot, GraphQL, Apache Kafka, Node js, Express
- **Data & Storage:** PostgreSQL, MongoDB, Amazon RDS, Hadoop, Elasticsearch, Redis, DynamoDB, RAG
- **Cloud & DevOps:** AWS (EC2 | S3 | Lambda | RDS), Docker, Terraform, GitHub Actions, Kibana

WORK EXPERIENCE

UT Dallas - International Center | *Technology & Operations Worker*

May 2025 - Present

- Acted as the central liaison between users, IT administrators, and support teams to coordinate onboarding and device provisioning, accelerating service delivery and reducing communication overhead.
- Managed end-to-end asset lifecycle workflows - imaging, access setup, deprovisioning, and issue resolution - standardizing processes to eliminate bottlenecks and improve operational efficiency.

Delta Systems | *Data Analyst*

Jan 2024 - Jul 2024

- Architected ETL pipelines with Python (Pandas, NumPy) and SQL to consolidate data from multiple structured sources into PostgreSQL, cutting processing time by 30% and increasing data reliability for downstream analytics
- Designed data transformation and validation workflows, adding schema normalization, data quality checks, and feature-engineering steps to create ML-ready datasets, which raised clustering accuracy and lowered noise in records
- Optimized SQL workloads by refactoring queries, adding indexes, and analyzing execution plans, which lowered query latency and allowed analytical reporting for business stakeholders
- Implemented k-means clustering with scikit-learn on customer data, creating segments that supported targeted marketing strategies and improved operational efficiency

PROJECTS

SignalTrace – Realtime News Analysis Software | [Code](#)

- Built a real-time entity intelligence pipeline using Kafka, PySpark Structured Streaming, Logstash, Elasticsearch, and Kibana to ingest live news, extract named entities, and surface rolling mention counts in a live dashboard.
- Designed an event-driven, schema-based architecture with decoupled Kafka topics for replayable ingestion and downstream analytics, enabling resilient stream processing and near-real-time search/visualization..

FinBERT Based Earnings Signal Engine | [Code](#)

- Built a RAG-augmented sentiment pipeline retrieving earnings transcripts, SEC filings, and news into a FinBERT scoring layer, applying entity resolution, source credibility weighting, and market residual modeling to isolate firm-level sentiment signals from retrieved financial context.
- Designed a weighted directed graph propagating cross-company sentiment spillover across supplier, competitor, and sector relationships; generated LLM-produced natural language explanations for next-day directional predictions, surfacing spillover attribution shares and top contributing firms per signal.

Traceline – Pipeline Anomaly Detection System | [Demo](#) | [Code](#)

(WINNER TIDAL HACK'26)

- Designed a feature-matching pipeline to link corrosion/anomaly records across inspections, improving match confidence and reducing manual review effort by 70%.
- Built a change-over-time analysis workflow to quantify corrosion progression and flag high-risk segments for prioritization using gradient boosting trees.